Anubhav Kumar Srivastava

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Professional summary

Quantum Control & Sensing Scientist at Ideaded S.L., Spain, working in optimal quantum control and quantum sensing targeting neutral-atom-based quantum computing.

7+ years of scientific programming experience developing quantum simulation tools and ML models; proficient in numerical techniques, including tensor networks, semidefinite programming, and variational circuit optimization.

Published researcher in top-tier journals with 100+ Google Scholar citations, complemented by a dedicated track record of public science outreach.

Technical Skills (Github link)

Programming: Python (advanced), Julia, Java, and C++

Quantum SDKs: Qiskit, PennyLane, QuSpin, QuTiP

Numerical & Optimization: Tensor networks (iTensor), semi-definite programming, convex optimization, quantum circuits

Machine learning: PyTorch, scikit-learn, attention-based neural networks, quantum machine learning

Scientific Computing: Large-scale simulation of many-body systems, variational algorithms, Hamiltonian learning

Professional Experience

Ideaded, Spain *2026 - current*
Quantum Control & Sensing Scientist
Research focus: Optimal quantum control for state preparation and gate implementation based on a neutral-atom platform. Quantum metrology and sensing using optimal control in many-body systems.

ICFO - The Institute of Photonic Sciences, Spain *2021 - 2026*
MSCA PhD fellow – Quantum Simulation of Spin Systems for Optimal Quantum Metrology
Supervisors: Prof. Dr. Maciej Lewenstein, Dr. Marcin Płodzień
Research focus: scalable simulation and certification of many-body quantum systems for optimal quantum metrology and algorithm benchmarking.
Impact: Awarded the Excellent grade with the highest *cum laude* distinction for the PhD thesis and public defense. Results published in leading journals, including Physical Review Research, Physical Review A, and Quantum.

Los Alamos National Laboratory, US *Jun - Aug 2024*
Summer research fellow (Quantum Computing Summer School 2024)
Research focus:
- Worked on thresholded magnetic field sensing with frustrated spin systems.
- Simulated nonlinear quantum sensing protocols.
- Collaborated in an interdisciplinary quantum computing and quantum error correction program.

Impact: Resulted in a publication in (*Physical Review Research 8, L012057 (2026)*).

Conferences & Talks

Invited Talks (1) : Quantum theory group (University of Geneva, 2024)

Contributed Talks (6): YQIS (Barcelona, 2025), FISMAT (Venice, 2025), ICFO-IMPRS workshop (Munich 2025, Barcelona 2023), ICE-9 (Tenerife, 2024), LANL QCSS (Los Alamos, 2024), YAO (Barcelona, 2023)

Poster presentations (9): Niels Bohr Quantum Summer School (Odense, 2026), XL Trobades (Mahón, 2025), Quantum matter out of equilibrium (Granada, 2025), CQD (Okinawa, 2024), QUMINOS (Les Diablerets, 2024), QIPA (Prayagraj, 2023), QTD (Vienna, 2023), Quantum simulation (Trieste, 2023), Quantum mixtures (Varenna, 2022)

Selected Projects & Publications (Google Scholar link)

Quantum Metrology Optimization Framework:
- Designed thresholded trimer quantum sensors for magnetometry. [*Phys. Rev. Research 8, L012057*]
- Implemented convex-optimization-based certification tools. [*Quantum 7, 1152 (2023)*]
- Designed scalable metrological performance estimators. [*arXiv.2603.24035 (2026)*]
- Applied to topological quantum thermometry models. [*Phys. Rev. A 111, 052216 (2025)*]
- Optimal quantum control framework for quantum sensing. [ongoing]
- Classical shadows-based quantum thermometry. [ongoing]

ML-Enhanced Quantum State Tomography:
- Built attention-based neural networks to reduce measurement overhead. [*Phys. Rev. Research 6, 033248 (2024)*]
- Benchmarked performance against conventional tomography methods.

Quantum Error Correction Modeling (ongoing projects):
- Simulated surface-code-inspired architectures.
- Studied GKP encoding performance under noise models.

Leadership & Community

- **Quantum Dequoded** (blog): Created and write *Quantum Dequoded*, a tech blog translating complex quantum physics and computing concepts into accessible insights for non-technical readers.
- **Frontiers for Young Minds (FYM)**: Accepted to contribute a research article, *Infinite multiverses and where to find them?*, to the special MSCA edition of FYM journal for school children in 230+ countries.
- **Co-supervised** Bachelor's and Master's students in quantum simulation and ML-based biophysics projects (2025-26).
- **ICFO Young Photonics**: As part of the outreach program at ICFO, **taught and mentored** school students on the basics of quantum mechanics and introduction to quantum computing in 2023 and 2025.
- **Co-founder** (2021) of QTPi, a quantum awareness hub with over 300+ Discord members and 10+ YT videos as part of the global outreach quantum lecture series.
- **Turing club**: **Co-founder** and former **co-convener** of the coding club at IISER Mohali.
- **Group coordinator**: Organizing weekly group meetings and scientific guest visits for the **QOT group** at **ICFO** (2022-25).
- **Workshops**: Regularly participated in **soft-skills workshops** on **public speaking**, **research ethics**, and **scientific communication**, organized by ICFO and BIST, Spain.

Languages

English (Native), Spanish (Intermediate), French (Beginner), Hindi (Native)

Other Information

Additional interests: Professionally trained in classical Indian guitar; active in quizzing and stand-up comedy; passionate about football and team sports.

References: Available upon request.